Started on	Monday, 10 March 2025, 2:58 PM
State	Finished
Completed on	Monday, 10 March 2025, 3:14 PM
Time taken	16 mins 7 secs
Grade	**80.00** out of 100.00

Question **1**

Correct

Mark 20.00 out of 20.00

Write a Python program to find the cube of all elements in a list using list comprehension

For example:

Input	Result
3 11.5 22 33.23	[11.5, 22.0, 33.23] [1520.875, 10648.0, 36693.65926699999]

Answer: (penalty regime: 0 %)

```
n=int(input())

lst=[float(input()) for _ in range(n)]
newlst=[i**3 for i in lst]

print(lst)
print(newlst)
```

	Input	Expected	Got	
✔	3 11.5 22 33.23	[11.5, 22.0, 33.23] [1520.875, 10648.0, 36693.65926699999]	[11.5, 22.0, 33.23] [1520.875, 10648.0, 36693.65926699999]	✔
✔	5 2 3.5 6 9 45	[2.0, 3.5, 6.0, 9.0, 45.0] [8.0, 42.875, 216.0, 729.0, 91125.0]	[2.0, 3.5, 6.0, 9.0, 45.0] [8.0, 42.875, 216.0, 729.0, 91125.0]	✔

Passed all tests! ✔

Correct

Marks for this submission: 20.00/20.00.

Question **2**

Correct

Mark 20.00 out of 20.00

Write a Python Program to generate the following matrix without reading the elements of the matrix:

For example:

Input	Result
5	Matrix: 5 0 0 0 0 0 5 0 0 0 0 0 5 0 0 0 0 0 5 0 0 0 0 0 5

Answer: (penalty regime: 0 %)

```
n=int(input())
print("Matrix:")
def mat(n):
    return [[n if i==j else 0 for j in range(n)]for i in range(n)]

def newmat(m):
    for i in m:
        print(*i)

m=mat(n)
newmat(m)
```

	Input	Expected	Got	
✔	5	Matrix: 5 0 0 0 0 0 5 0 0 0 0 0 5 0 0 0 0 0 5 0 0 0 0 0 5	Matrix: 5 0 0 0 0 0 5 0 0 0 0 0 5 0 0 0 0 0 5 0 0 0 0 0 5	✔
✔	4	Matrix: 4 0 0 0 0 4 0 0 0 0 4 0 0 0 0 4	Matrix: 4 0 0 0 0 4 0 0 0 0 4 0 0 0 0 4	✔

Passed all tests! ✔

Correct

Marks for this submission: 20.00/20.00.

Question **3**

Not answered

Mark 0.00 out of 20.00

Given an array arr[] of size n, its prefix sum array is another array prefixSum[] of the same size,

such that the value of prefixSum[i] is arr[0] + arr[1] + arr[2] … arr[i]. Write a Python code to generate the prefixSum []

Input : arr[] = {10, 20, 10, 5, 15}
Output : prefixSum[] = {10, 30, 40, 45, 60}

For example:

Test	Input	Result

`n = int(input())`
`arr=createList(n)`
`prefix=fillPrefixSum(arr)`
`print(arr)`
`print(prefix)`
	3
	11
	22
	33
		[11, 22, 33]
		[11, 33, 66]

Answer: (penalty regime: 0 %)

```

```

Question 4

Correct

Mark 20.00 out of 20.00

Write a python program to check whether the number '41' is greater or equal or lesser than the number '29' or not

For example:

Input	Result
41 29	41 is greater than 29

Answer: (penalty regime: 0 %)

```
a=int(input())

b=int(input())

if a>b:
    print(f"{a} is greater than {b}")

else:
    print(f"{a} is smaller than {b}")

```

	Input	Expected	Got	
✔	41 29	41 is greater than 29	41 is greater than 29	✔
✔	89 100	89 is smaller than 100	89 is smaller than 100	✔
✔	12 23	12 is smaller than 23	12 is smaller than 23	✔

Passed all tests! ✔

Correct

Marks for this submission: 20.00/20.00.

Question **5**

Correct

Mark 20.00 out of 20.00

Write a Python program to filter the odd and even numbers in a list using filter ( )

For example:

Input	Result
5 34 57 89 24 11	[34, 24] [57, 89, 11]

Answer: (penalty regime: 0 %)

```
n=int(input())

lst=[int(input()) for _ in range(n)]

oddno=list(filter(lambda x:x%2==0,lst))
evenno=list(filter(lambda x:x%2!=0,lst))

print(oddno)
print(evenno)
```

	Input	Expected	Got	
✔	5 34 57 89 24 11	[34, 24] [57, 89, 11]	[34, 24] [57, 89, 11]	✔

Passed all tests! ✔

Correct

Marks for this submission: 20.00/20.00.